Oscar Jakacki

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EDUCATION

Northwestern University, Evanston, IL — Sept 2025 - Expected Jun 2028
B.S. Electrical Engineering Coursework: Advanced Digital Design, Fundamentals of Circuits

Temple University, Philadelphia, PA — Aug 2024 - May 2025
Bioengineering

EXPERIENCE

Silicon/RF Intern, Stealth Hardware Startup — Jun 2026 - Sept 2026
San Diego, CA

- Built Python tooling for hardware bring-up testing, including a GPS post-processing pipeline with signal filtering that brought location accuracy to within 20 m.
- Wrote a software spectrum analyzer running on in-house hardware, used for RF interference analysis in place of dedicated bench equipment.
- Worked on FPGA configuration and Linux-based test infrastructure.

Undergraduate Researcher, Embodied Systems Lab — Mar 2026 - Present
Northwestern University, Evanston, IL

- Designed op-amp signal conditioning circuits for a biosignal acquisition system built into smart glasses, detecting heart rate and facial activity from wearable sensors.
- Designed and built the signal conditioning PCB from scratch; develop and debug embedded C firmware on a XIAO ESP32.

Electrical Team Member, Northwestern Solar Car (NUsolar) — Jan 2026 - Present
Evanston, IL

- Redesigned the battery management system with a teammate, moving to a passive cell balancing architecture; improved cell voltage uniformity and system efficiency. Schematic and PCB work in Altium Designer.

Research Assistant, Querrey Simpson Institute for Bioelectronics — Sept 2025 - Jan 2026
Northwestern University, Evanston, IL

- Improved PCB and enclosure designs in Altium Designer and Fusion 360 for wearable devices collecting vitals from newly transplanted tissue; collected and processed sample vitals data from human subjects to support device iteration.

Undergraduate Researcher, Pleshko Lab — Dec 2024 - Sept 2025
Temple University, Philadelphia, PA

- Studied how sample orientation affects O-PTIR spectroscopy of bone, enabling comparison of O-PTIR data against conventional FTIR; collected and analyzed spectroscopic data on bone samples following drug treatment.

PROJECTS

Real-Time Music Genre Classifier on FPGA — Verilog, Spartan-7, Vivado

- Built an audio classification pipeline entirely in hand-written Verilog with no processor and no vendor IP: 1024-point FFT, 26-band mel filterbank, and a quantized neural network, fixed-point throughout.
- 86.3% accuracy on 3,192 held-out samples and 67% on live audio played into the board; met timing with 71.4 ns slack and zero failing endpoints.
- Traced a persistent misclassification bug to a clock constraint mapped to the wrong oscillator, which neither simulation nor timing analysis could detect.

Web Server on a Disposable Vape Microcontroller — C, Cortex-M0+, OpenOCD

- Served HTTP from a PY32F030 (32 KB flash, 4 KB RAM) with no network hardware, tunnelling TCP/IP through the chip's SWD debug port using ARM semihosting and SLIP.
- Designed a breakout PCB around the chip and had it fabricated after the original board's pads proved unreliable to solder.
- Diagnosed and patched a defect in OpenOCD that discarded inbound semihosting data whenever the target was not already blocked in a read.

Attenda Health - Grand Prize, Second Place Overall, WildHacks 2026 — Next.js, React, Supabase

- Proposed and helped build a mobile app that decentralizes hospital call bells, based on observations while volunteering at a hospital; placed 2nd of 68 submissions.

PUBLICATIONS

Dave, R. N., Prodanich, J., Lai, Y., **Jakacki, O.**, Lin, S., Thoene, J., Alshurafa, N., & Arora, N. (2026). *GlassTENG: Self-Powered Triboelectric Nanogenerator based Sensing of Pulse, Jaw, and Upper Facial Activity from Everyday Glasses.* ISWC 2026, Notes and Briefs. — 2026

Jakacki, O., Kust, S., & Pleshko, N. (2025). *Characterization of O-PTIR Orientation Effects on Bone.* Poster, Sci-X Conference. First author. — Oct 2025

SKILLS

Languages	Verilog, C, Python, Bash
Digital / DSP	RTL design and simulation, FPGA bring-up, timing constraints, Vivado; FFT, digital filtering, fixed-point arithmetic
Hardware	PCB design (Altium Designer, KiCad, Fusion 360), op-amp signal conditioning, soldering and board rework; RF test equipment, oscilloscopes, logic analyzers
Platforms	ARM Cortex-M, ESP32, Spartan-7 FPGA, Raspberry Pi, Linux, Git